Digital Forensics: Focusing on Image Forensics

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***Abstract*— The world is continuously developing, and people's needs are increasing as well; so too are the number of thieves increasing, especially electronic thieves. For that reason, companies and individuals are always searching for experts who will protect them from thieves, and these experts are called digital investigators. Digital forensics has a number of branches and different parts, and image forensics is one of them. The budget for the images branch goes up every day in response to the need. In this paper we offer some information about images and image forensics, image components and how they are stored in digital devices and how they can be deleted and recovered. We offer general information about digital forensics, focusing on image forensics.**

Keywords—image, forensics, investigations, processing, digital, cyber

I. INTRODUCTION

Nowadays, all around the world, electronic crime is increasing, making it vital to enhance all methods that combat this type of crime. Digital forensics is one important way to investigate electronic crime.

Technology advances every day. The difference between digital technology and other types of technology is that digital technology develops faster. We can feel this fast development when we browse the Internet. Every hour there are numerous reports and researchers talking about new inventions, new programs, new technology, or new devices, etc. that aim to serve our society — hence, the problems associated with this rapid development demand a corresponding rapid response. Digital forensics is that response.

Because of the huge number of readily available software programs that can manipulate digital media (especially images and video), it is necessary to enhance digital forensics continuously to make it able to detect the manipulation of data, protect data from attackers, and recover deleted evidence.

Image is one of the more popular media for publishing data or news (political, art, social news), so it is important to make sure that we can verify whether this type of data is real or fake, who is authorized to publish it, and who is behind the spread of fake news.

Image forensics is the branch that takes care of this matter. It uses various methods to process images in order to detect evidence, recover deleted evidence, and make suitable reports about events to submit to court or to responsible entities.

II. WHAT IS A DIGITAL IMAGE AND WHAT IS IT COMPOSED OF?

A digital image is an array of pixels arranged in a certain way, showing a specific image. Pixels are the smallest unit in the image (the name is derived from the picture element) [1].

We store images or transfer them from one place to another inside the computer or transfer them to another memory like flash memory as an array of bits (0, 1) [1]. When we talk about an image, we should refer to its types. Images are divided mainly into two types:

- pixel images
- vector images.

A. *Pixel images (bitmap images)* — We deal with these every day (normal digital images). PNG, JPEG, TIFF, and GIF are all pixel images. Each type is defined by its own programs and devices. They are pixel-based; size is linked to its details; they are best for editing and manipulating and are not scalable (Table 1) [1].

B. *Vector images* — These images are used for cartoons, logos, descriptions, etc. and are used by special programs like Corel-draw with a mathematical calculation. When we do operations like enlarging or stretching, the resolution of the image will stay the same. Types of vector images are AI, CDR, SVG, and EPS [1].

TABLE 1. COMMON IMAGE FORMATS [2, 3]

Acronym	Full name
JPEG	Joint photograph experts group
BMP	Bitmap picture
GIF	Graphics interchange format
PNG	Portable network graphics
TIFF	Tagged image file format
PSD	Photoshop document
XCF	Experimental Computing Facility
AI	Adobe illustrator artwork
CDR	Corel draw image

An image is an array of pixels, as mentioned above, but there are several image types [2]:

- binary image
- gray scale image
- colored image
- multispectral image.

A. A *binary image* is an array of two dimensions of bits. It has only two colors, typically white and black — 0s represent white-colored pixels, and 1s represent black-colored pixels [2].

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B. *Gray scale images* are those images where each pixel is represented with two bits, representing the power of the white/black colors (00 represents white color, 11 represents black color) and the values between these two limits represent the power of the gray color [2].

C. *Colored images,* or what are known as RGB images, (red, green, blue), are the images that contain a variety of colors [2]. This type of image is represented inside the memory by a byte for each pixel. Each pixel contains an amount of red, green, blue power, and is stored as bits:

Red green blue
00 00 00

D. *Multispectral images* are images consisting of multi-layers of colors, with each layer viewing specific colors (red, green, blue) [2].

III. Storing an Image and Deleting It

Images, like all other data, are stored in a computer, so it is important to explain briefly the data types. Firstly, data are divided into three types: active, latent, and archival. This classification makes it easier to file systems when searching data [6].

A. *Active data:* the data used daily and readily available [6].

B. *Latent data:* data not accessed from the operating system for a long time but still stored inside the memory and can be accessed by forensic tools [6].

C. *Archival data*: historical data available on hard disk or flash memory [6].

The most important information about deleted files is that when users delete data the operating system allocates that location as free location, and it still stores the data but without making it accessible. When will this data be erased? When the user stores more data, the operating system overwrites the new data on the old data [6]. From this point digital forensics begins.

IV. File System

The paragraph above mentioned how the operating system stores important information including files in common use and images. We will explain this topic further [6].

Firstly, file systems are divided into three main types:

- FAT (allocation file system)
- NTFS (new technology file system
- HFS+ (hierarchical file system).

When new data are overwritten on the old data, the old data will be unrecoverable, except that data that we call slack space.

V. Slack Space

As mentioned above, when a user deletes data, the operating system allocates that space of memory as empty space or unallocated space and the data remain till the user stores more data and the operating system overwrites the new data on the old data [6]. Sometimes when the piece of new overwritten data is smaller than the old data, the memory stores an amount of old data too, and this is called slack space — hence, digital forensic tools image these data and process them to check if they hold any important information [6].

VI. Digital Forensics

Digital forensics is the participation of computer science and digital technology in investigation procedures to gather digital evidence of a crime. Using specialized tools and mathematical equations, it is equivalent to physical evidence of crime detection, and it is presented in the form of a report that can be used as evidence in court [6].

In an attempt to study all the content to find evidence that will convince the court [6], it is up to digital investigations to find the appropriate solutions by searching all electronic devices located at the scene of the crime or incident — from computers, laptops, mobile phones, internal networks, Internet and e-mail, and beyond — to analyze video, audio and image.

This area has been used extensively to include investigation of crimes, accidents, administrative matters, civil litigation, cyber penetration, and intelligence. Scientific institutions are constantly developing this area because of its necessity and importance in the service of justice.

VII. Digital Forensics Tools

The tools and websites that investigators use in their investigations are vital because they represent the heart of digital forensics. Because of that, we will explain the main digital forensics tools.

Figure 1. Visual time analyzer [11]

A. *Visual time analyzer* is a program that gives the user full control of a target computer. It presents full details about all working programs on the computer, such as the time users spend using each program or the Internet; it also tracks working time, pauses etc. For that reason,

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sometimes parents use it to control their children's use of the computer [7]. It supports users with information on software metering, Internet use, computer supervision, project overview etc. Below is an image of this program [7].

B. *X-ways forensics* is a program, based on WinHex, that is designed to work as an advanced forensics program. It checks image files and all directories, if the directory is not a continuous segment, and presents the information to the investigator. It supports multiple types of files such as FAT, NTFS, CDFS, UDF and others. Below is an image of the program [7].

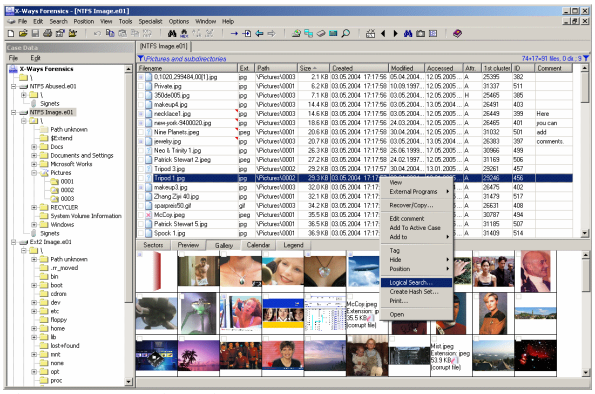

Figure 2. X-ways forensics [11]

C. *Evidor* is a program mostly used with text documents. It takes a specific keyword and begins searching on it inside the memory. It will retrieve all data that contain this keyword and show it to an investigator [7].

VIII. Image Forensics

As mentioned above about images and how they are stored and deleted, in the image forensics branch digital investigators should be familiar with dealing with images stored on hard drives, flash memories, CDS, DVDS, and how they can be extracted or recovered. In this part we will discuss how and when a digital investigator can deal with images. But firstly, here is some further information about images [7]:

A. *Image file header* — In every file stored inside the computer there is a part of it that contains information about it. For example, in image files the header contains image color number, image size, the resolution, and some other data. It is used by programs to define the image — so, if this file is missing, programs cannot read the image [7].

B. *File fragments* — As we showed above, if an image is deleted there is a possibility it can be reconstructed by taking the slack space information and trying to read it through special programs like Hex Editor or another program [7]. After extracting the slack parts, the investigator will get some information about the contents and may be able to predict the missing parts [7].

C. *Unknown files* — When an investigator extracts files from a computer or a hard disk, there will possibly be some unknown files (the extension is unknown or missing). The investigator cannot determine what kind of file it is [7]. In this state, the investigator needs a special tool to define the file. One of these tools is a website known as FILExt (www.filext.com). This website will compare the file with its own online database, and then it will give the following information about the file [7]:

1) the file extension

2) information about the file — whether it is a document or image or whatever and a suitable program to open it

3) the file ownership — this file belongs to which company or institution

4) possibly a link to a program that can be used to open this file.

IX. Using Digital Forensic Tools to Deal with Images

The aim of image digital forensics is to extract images and deal with them, so we will discuss some of the tools that deal with images.

A. *GFE (Graphic File Extractor)* is a program that uses the DOS system, which copies discs, Windows system, UNIX, and all programs compatible with INTL. It is designed to work on disk files and make a second copy using Safeback bitstream, which can retrieve memory files and make a backup, whatever the size of the memory. It is important that the program recovers files without any compression. This program is available on www.forensics-intl.com [7].

B. *P2 eXplorer* is a program used to view images that are extracted with the Safeback program; the investigator can use it to discover image-retrieved content. This program is available on www.paraben-forensics.com [7].

C. *ILOOK* is a program that provides the investigator with a graphical user interface similar to Windows Explorer. The investigator can explore the suspected computer, extract what is needed from the system, and then explore it on another program or on the same program that provides a HEX viewer. It is available on www.ilook-forensics.org [7].

X. Photogrammetry Forensics

This is the method or system followed in digital investigations. It involves taking a picture of the scene or target person and turning it into a three-dimensional graphic or drawing. This method is followed by digital investigators when it is necessary to construct a three-dimensional design to know more details about a crime or accident, or when asked about the position of the targeted person or the perpetrator [4].

Sometimes it is taken from one angle and at other times pictures are taken from more than one angle to enable the

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program to do a thorough survey and statement of most corners of the incident [4].

When two or more images are taken of a place or person, it makes it easier to determine everything's position and size by crossing the points of multiple cameras. This is known as multi-image photogrammetry [4].

There are two types of images:

- controlled image photogrammetry
- uncontrolled image photogrammetry.

Photogrammetry using photos taken especially for the investigation is called controlled image photogrammetry, while when investigators use images that are taken accidentally, this is called uncontrolled image photogrammetry [4].

Regarding methods of detecting in photogrammetry, there are three types:

- ortho-rectification
- space resection and multi-image photogrammetry
- reverse projection.

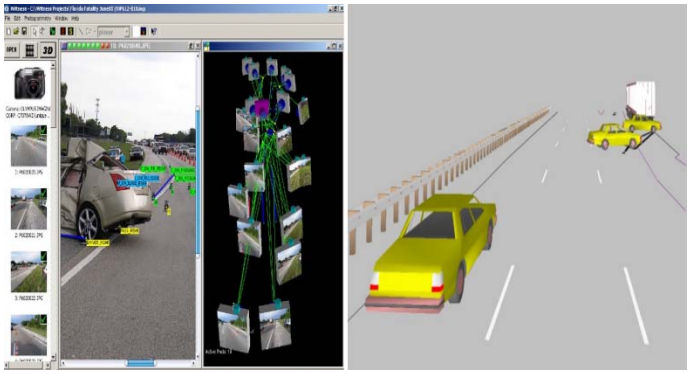

Figure 3. Photogrammetry forensics [5]

XI. DIGITAL FORENSIC SURVEYS

Finally, we will feature some of the categories of digital forensics around the world.

A. Digital security's future — Every branch proves its important when it expands continuously. According to an American website, which has scouted U.S. technology growth expected in 2019, cybersecurity concerns are level with economic growth [10].

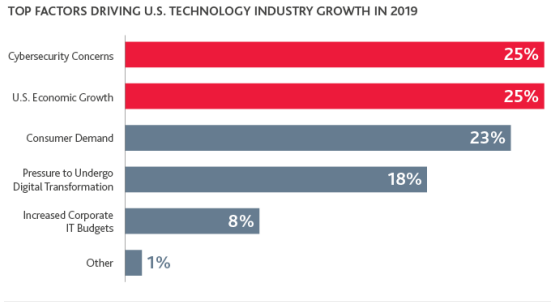

Figure 4. Top factors driving U.S. technology industry growth in 2019 [10]

B. Computer crimes — If we follow digital forensics, we will find that computer crimes are growing rapidly, unfortunately. Figure 5 gives the results of a survey about computer crimes [9].

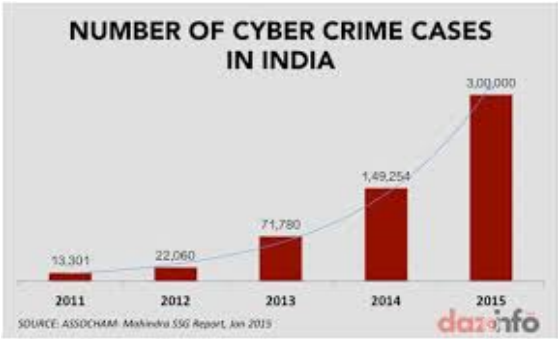

Figure 5. Computer crimes growing [9]

C. Digital forensics budget — In every branch, including digital forensics, the salary amount reflects the importance of that branch. When we observe budget and salaries assigned to the digital forensics branch, we observe the importance of it. Figure 6 shows the expected development in the digital forensics budget [8].

D.

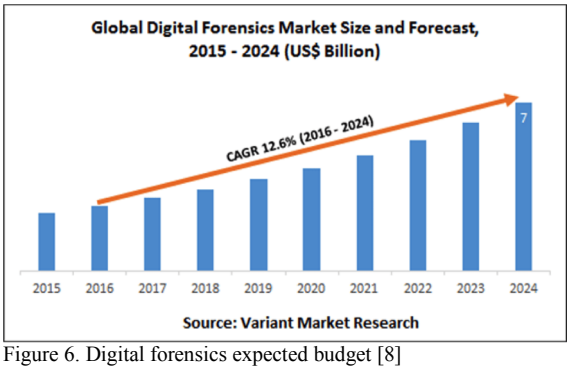

Figure 6. Digital forensics expected budget [8]

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XII. CONCLUSION

Each day, digital crimes expand and develop more and more. This is why people are losing confidence in the digital or virtual world. All programs, video, sound, image, even computers and mobile phones have become the target of penetration and electronic manipulation. We have highlighted some of the features of the digital forensic specialty, focusing on images, as this paper was interested in explaining the image and how it is stored on digital devices and how it is deleted, and what the specific activation of operating systems is when it comes to deleting an image. We showed that some information remains on the hard disk after deleting the data, and how to retrieve this information. We also highlighted some features of digital forensics, forensic digital images, and some uses for digital forensic images.

Finally, we wanted to help interested persons to keep abreast of the latest important statistics on digital forensics.

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